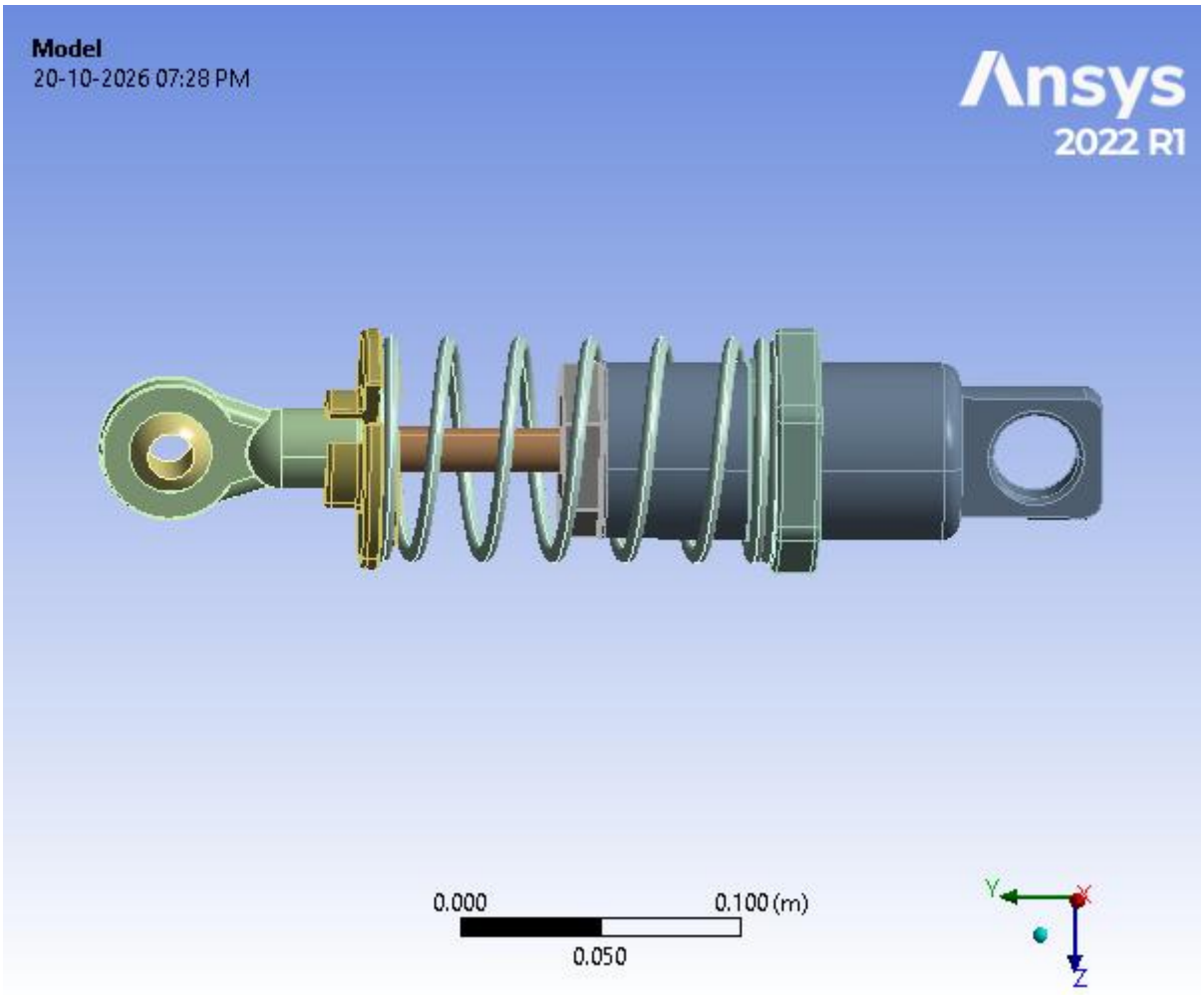
Shock Absorber Analysis



Contents

- **Units**
- **Model (A4)**
  - Geometry Imports
    - Geometry Import (A3)
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    - Loads
    - Solution (A6)
      - Solution Information
      - Results
      - Stress Tool
        - Safety Factor
      - Fatigue Tool
        - Results
- **Material Data**
  - Aluminum Alloy 6061-T6
  - Low Alloy Steel 4140 - Hardened and Tempered
  - Carbon Steel 1095 - Hardened and Tempered
  - ASTM A228 Music Wire
  - Rubber, nitrile (NBR)

Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved

Definition	
Source	D:\Solidworks\Suspension\Project - Shock Absorber\STEP - Real Shock Absorber.STEP
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Attribute Key	SDFEA;DDM
Named Selections	No
Named Selection Key	NS
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Compare Parts Tolerance	Tight
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Stitch Tolerance	0.0000001
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	D:\MNNIT Allahabad\Solidworks\Complete-Vehicle-Suspension-System\Ansys_Files\Shock Absorber Analysis\Shock Absorber Analysis\Shock Absorber Analysis_files\import_files\STEP - Real Shock Absorber.STEP
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	8.5725e-002 m
Length Y	0.35849 m

Length Z	8.8703e-002 m
Properties	
Volume	4.8589e-004 m³
Mass	1.9318 kg
Scale Factor Value	1.
Statistics	
Bodies	15
Active Bodies	13
Nodes	152437
Elements	81618
Mesh Metric	Element Quality
Min	3.1558e-002
Max	1.
Average	0.78395
Standard Deviation	0.15834
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	Program Tolerance
Decompose Disjoint Geometry	Yes

Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

ck 2]	*STEP - Real Shock Absorber-FreeParts|Scale1[13]*	*STEP - Real Shock Absorber-FreeParts|Scale1[11]*	*STEP - Real Shock Absorber-FreeParts|Scale1[1]*	*STEP - Real Shock Absorber-FreeParts|Scale1[4]*	*STEP - Real Shock Absorber-FreeParts|Scale1[6]*	*STEP - Real Shock Absorber-FreeParts|Scale1[9]*	*S Fre*
Meshed							
Graphics Properties							
Yes							
1							
Definition							
No							
Flexible							
Default Coordinate System							
By Environment							
None							
Material							
	Low Alloy Steel 4140 - Hardened and Tempered	Carbon Steel 1095 - Hardened and Tempered	ASTM A228 Music Wire	Aluminum Alloy 6061-T6	Carbon Steel 1095 - Hardened and Tempered		
Yes							
Yes							
Bounding Box							
3.175e-002 m			8.0791e-002 m	3.302e-002 m	5.207e-002 m	3.175e-002 m	
	9.366e-002 m	1.905e-003 m	0.1519 m	3.302e-002 m	8.3382e-003 m	1.905e-003 m	
	5.1137e-002 m	3.175e-002 m	8.0772e-002 m	3.302e-002 m	5.207e-002 m	3.175e-002 m	
Properties							
	4.1971e-005 m³	7.6064e-007 m³	4.071e-005 m³	1.2114e-005 m³	7.152e-006 m³	7.6064e-007 m³	
	0.32947 kg	5.971e-003 kg	0.31957 kg	3.2865e-002 kg	1.9403e-002 kg	5.971e-003 kg	
	5.4237e-008 m	-1.1714e-003 m	-2.9517e-005 m	2.1522e-018 m	1.3165e-007 m	-1.1714e-003 m	
	0.12047 m	-7.5502e-002 m	-1.7526e-003 m	0.14649 m	-4.0228e-002 m	-6.169e-002 m	

	1.2247e-007 m	-1.2513e-003 m	3.4143e-004 m	2.0021e-019 m	-6.1781e-006 m	-1.2513e-003 m	
²	2.7409e-004 kg·m²	4.53e-007 kg·m²	9.8607e-004 kg·m²	4.8864e-006 kg·m²	4.4699e-006 kg·m²	4.53e-007 kg·m²	9.
²	5.433e-005 kg·m²	9.878e-007 kg·m²	4.5312e-004 kg·m²	3.7804e-006 kg·m²	8.7214e-006 kg·m²	9.878e-007 kg·m²	1.
²	2.4339e-004 kg·m²	5.3842e-007 kg·m²	9.8606e-004 kg·m²	3.7804e-006 kg·m²	4.4696e-006 kg·m²	5.3842e-007 kg·m²	9.
Statistics							
	4082	5697	8690	6655	6231	5705	
	2268	920	3079	4067	3169	920	
Element Quality							
	3.1558e-002	0.36815	3.7436e-002	0.43931	0.14491	0.36797	
	0.99936	1.	0.99992	0.99996	0.99995	0.99999	
	0.56504	0.94784	0.73022	0.82457	0.71852	0.94898	
	0.27111	6.5099e-002	0.24704	0.10047	0.17116	6.8541e-002	

TABLE 6
Model (A4) > Geometry > Parts

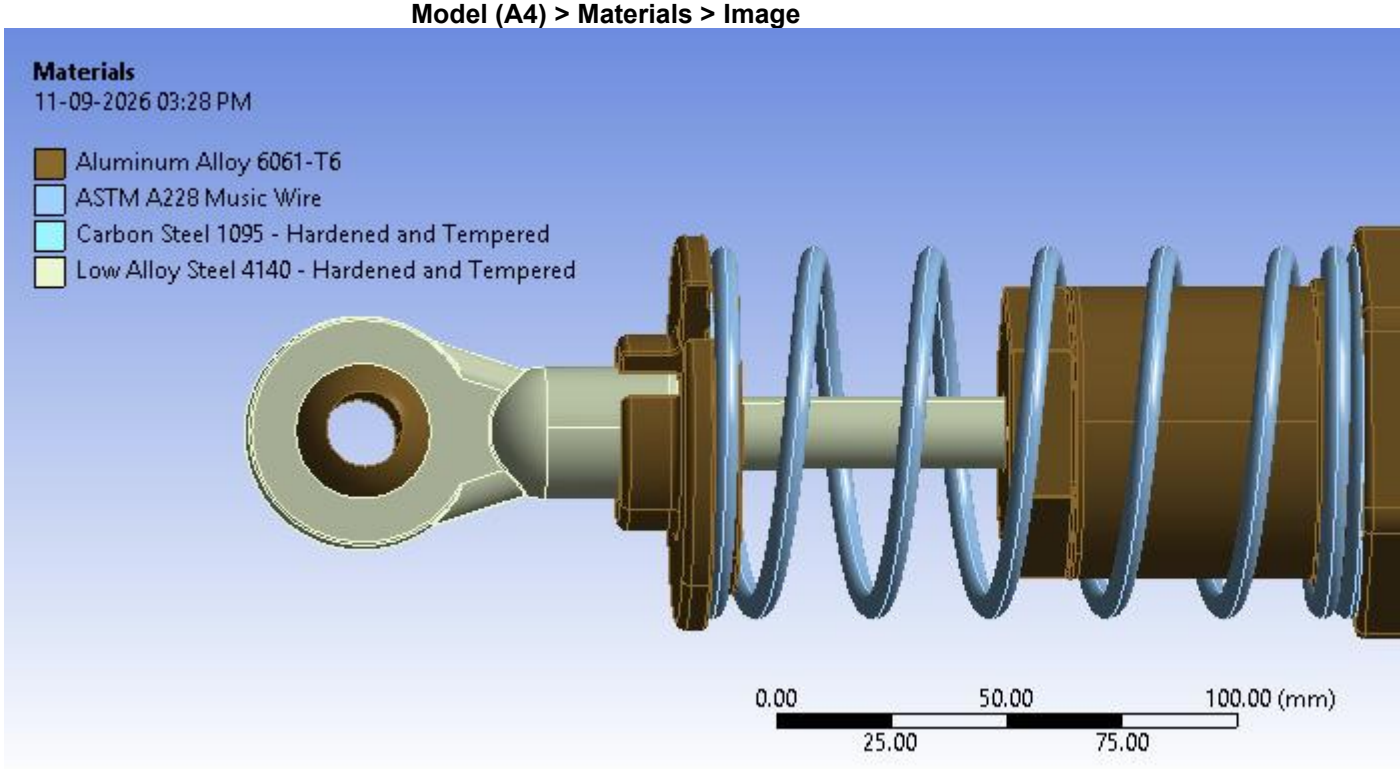
Object Name	*STEP - Real Shock Absorber-FreeParts|Scale1[8]*	*STEP - Real Shock Absorber-FreeParts|Scale1[5]*	*STEP - Real Shock Absorber-FreeParts|Scale1[3]*	*STEP - Real Shock Absorber-FreeParts|6. Assembled Shock Body-1/5. Assembled Shock Cartridge-1/7. O-Ring-2-solid1*
State	Meshed			Suppressed
Graphics Properties				
Visible	Yes			No
Transparency	1			
Definition				
Suppressed	No			Yes
Stiffness Behavior	Flexible			
Coordinate System	Default Coordinate System			
Reference Temperature	By Environment			
Treatment	None			
Material				
Assignment	Low Alloy Steel 4140 - Hardened and Tempered	Aluminum Alloy 6061-T6		Rubber, nitrile (NBR)
Nonlinear Effects	Yes			
Thermal Strain Effects	Yes			
Bounding Box				
Length X	1.5875e-002 m	8.4665e-002 m	8.5725e-002 m	6.3627e-003 m
Length Y	0.18732 m	2.54e-002 m	2.542e-002 m	1.5875e-003 m
Length Z	1.5875e-002 m	8.8703e-002 m	8.5725e-002 m	6.3627e-003 m
Properties				

Volume	3.5251e-005 m³	3.8888e-005 m³	4.7997e-005 m³	2.9186e-008 m³
Mass	0.27672 kg	0.1055 kg	0.13022 kg	2.8731e-005 kg
Centroid X	1.2289e-013 m	1.976e-005 m	-1.986e-003 m	1.0329e-007 m
Centroid Y	1.1027e-002 m	-7.6008e-002 m	7.5674e-002 m	-7.2212e-003 m
Centroid Z	-1.0457e-013 m	2.0406e-005 m	9.7499e-004 m	1.8202e-007 m
Moment of Inertia Ip1	7.7094e-004 kg·m²	7.471e-005 kg·m²	6.3304e-005 kg·m²	9.283e-011 kg·m²
Moment of Inertia Ip2	8.3523e-006 kg·m²	1.4283e-004 kg·m²	1.1114e-004 kg·m²	1.7663e-010 kg·m²
Moment of Inertia Ip3	7.7094e-004 kg·m²	7.4721e-005 kg·m²	5.4629e-005 kg·m²	9.2715e-011 kg·m²
Statistics				
Nodes	2313	23006	30386	0
Elements	1174	13877	18654	0
Mesh Metric	Element Quality			
Min	0.10758	6.4314e-002	3.2891e-002	0.
Max	0.99986	0.99951	0.99949	0.
Average	0.72322	0.79261	0.77449	0.
Standard Deviation	0.24812	0.13084	0.15243	0.

TABLE 7
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	6
Material Assignments	0

FIGURE 1
Model (A4) > Materials > Image



Coordinate Systems

TABLE 8
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. m
Origin Y	0. m
Origin Z	0. m
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Connections

TABLE 9
Model (A4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes

TABLE 10
Model (A4) > Connections > Contacts

Object Name	*Contacts*
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	9.4781e-004 m
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies

Statistics	
Connections	19
Active Connections	19

TABLE 11
Model (A4) > Connections > Contacts > Contact Regions

egion 2	*Frictional - STEP - Real Shock Absorber-FreeParts|Scale1[7] To STEP - Real Shock Absorber-FreeParts|Scale1[12]*	*Contact Region 4*	*Contact Region 5*	*Contact Region 6*	*Contact Region 7*	*Contact Region 8*	*Contact Regio*
Fully Defined							
Scope							
Geometry Selection							
	6 Faces	2 Faces	6 Faces		3 Faces	5 Faces	2 Faces
	2 Faces				3 Faces	14 Faces	2 Faces
hock Absorber-FreeParts|Scale1[7]			STEP - Real Shock Absorber-FreeParts|Scale1[2]		STEP - Real Shock Absorber-FreeParts|Scale1[13]		STEP - Real Sh Absorber-FreeParts|Scale
al Shock ber-Scale1[9]	STEP - Real Shock Absorber-FreeParts|Scale1[12]	STEP - Real Shock Absorber-FreeParts|Scale1[8]	STEP - Real Shock Absorber-FreeParts|Scale1[10]	STEP - Real Shock Absorber-FreeParts|Scale1[8]	STEP - Real Shock Absorber-FreeParts|Scale1[3]		STEP - Real Sh Absorber-FreeParts|Scale
No							
Definition							
	Frictional	Bonded					
Automatic						Manual	Automatic
Program Controlled							
Program Controlled							
9.4781e-004 m							9.4781e-004
No							
	0.2						
Advanced							
Program Controlled							
Program Controlled							
Program Controlled							
Program Controlled							
Program Controlled							
Program Controlled							
Program Controlled							
	Radius	Program Controlled					

	0.	
	2.e-003 m	
	None	

Geometric Modification
None
None

	Add Offset, No Ramping	
	0. m	

TABLE 12
Model (A4) > Connections > Contacts > Contact Regions

ct Region 12	*Contact Region 13*	*Contact Region 14*	*Contact Region 15*	*Frictional - STEP - Real Shock Absorber-FreeParts|Scale1[10] To STEP - Real Shock Absorber-FreeParts|Scale1[8]*	*Contact Region 17*	*Contact Region 18*	
Fully Defined							
Scope							
Geometry Selection							
0 Faces	4 Faces	7 Faces	5 Faces	2 Faces			
8 Faces	2 Faces		3 Faces	2 Faces	4 Faces	3 Faces	
- Real Shock Absorber-reeParts|Scale1[6]	STEP - Real Shock Absorber-FreeParts|Scale1[9]		STEP - Real Shock Absorber-FreeParts|Scale1[10]		STEP - Real Shock Absorber-FreeParts|Scale1[12]	STEP - Real Shock Absorber-FreeParts|Scale1[8]	
- Real Shock Absorber-arts|Scale1[10]	STEP - Real Shock Absorber-FreeParts|Scale1[8]		STEP - Real Shock Absorber-FreeParts|Scale1[12]	STEP - Real Shock Absorber-FreeParts|Scale1[8]	STEP - Real Shock Absorber-FreeParts|Scale1[5]	Multiple	
No							
Definition							
Bonded				Frictional		Bonded	
Automatic	Manual		Automatic			Manu	
Program Controlled							
Program Controlled							
9.4781e-004 m			9.4781e-004 m				
No							
				0.2			

Advanced		
Program Controlled		
Program Controlled		
Program Controlled		
Program Controlled		
Program Controlled		
Program Controlled		
Program Controlled		
Program Controlled	Radius	Program Controlled
	0.	
	3.e-003 m	
	None	
Geometric Modification		
None		
None		
	Add Offset, No Ramping	
	0. m	

Mesh

TABLE 13
Model (A4) > Mesh

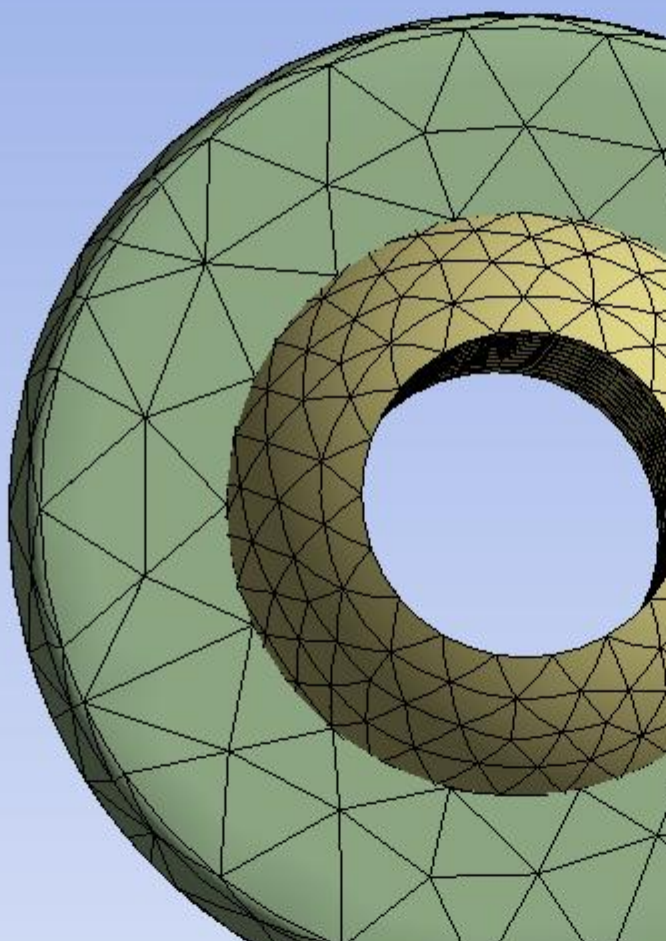
Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	3.e-003 m
Sizing	
Use Adaptive Sizing	No
Growth Rate	1.2
Max Size	1.e-002 m
Mesh Defeaturing	Yes
Defeature Size	Default (1.5e-005 m)
Capture Curvature	No
Capture Proximity	No

Bounding Box Diagonal	0.37912 m
Average Surface Area	3.6576e-004 m²
Minimum Edge Length	5.5414e-005 m
Quality	
Check Mesh Quality	Yes, Errors and Warnings
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	Element Quality
Min	3.1558e-002
Max	1.
Average	0.78395
Standard Deviation	0.15834
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Default (2.7e-005 m)
Generate Pinch on Refresh	No
Statistics	
Nodes	152437
Elements	81618

TABLE 14
Model (A4) > Mesh > Mesh Controls

Object Name	*Body Sizing*	*Body Sizing 2*	*Body Sizing 3*	*Body Sizing 4*
State	Suppressed	Fully Defined		
Scope				
Scoping Method	Geometry Selection			
Geometry	No Selection	2 Bodies	1 Body	3 Bodies
Definition				
Suppressed	No			
Active	No, Suppressed Geometry			
Type	Element Size			
Element Size	5.e-004 m	1.e-003 m	5.e-003 m	7.e-003 m
Advanced				
Defeature Size	Default (1.5e-005 m)			
Behavior	Soft			
Growth Rate	Default (1.2)			
Capture Curvature	No			
Capture Proximity	No			

FIGURE 2
Model (A4) > Mesh > Image



Named Selections

TABLE 15
Model (A4) > Named Selections > Named Selections

Object Name	*Selection*
State	Suppressed
Scope	
Scoping Method	Worksheet
Geometry	No Selection
Definition	
Send to Solver	Yes
Visible	Yes
Program Controlled Inflation	Exclude
Statistics	
Type	Manual
Total Selection	1 Node
Suppressed	1
Used by Mesh Worksheet	No
Tolerance	
Tolerance Type	Program Controlled
Zero Tolerance	1.e-008
Relative Tolerance	1.e-003
Angular Tolerance	1. °

Static Structural (A5)

TABLE 16
Model (A4) > Analysis

Object Name	*Static Structural (A5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 17
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	
Number Of Steps	3.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	1.

Minimum Substeps	1.
Maximum Substeps	10.
Solver Controls	
Solver Type	Direct
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Off
Output Controls	
Stress	Yes
Surface Stress	No
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No

Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\MNNIT Allahabad\Solidworks\Complete-Vehicle-Suspension-System\Ansys_Files\Shock Absorber Analysis\Shock Absorber Analysis\Shock Absorber Analysis_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	mks

TABLE 18
Model (A4) > Static Structural (A5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time	Auto Time Stepping	Define By	Initial Substeps	Minimum Substeps	Maximum Substeps
1	1. s	On	Substeps	1.	1.	10.
2	2. s	Program Controlled				
3	3. s					

FIGURE 3
Model (A4) > Static Structural (A5) > Image

A: Static Structural

Cylindrical Support

Time: 1. s

11-09-2026 03:36 PM

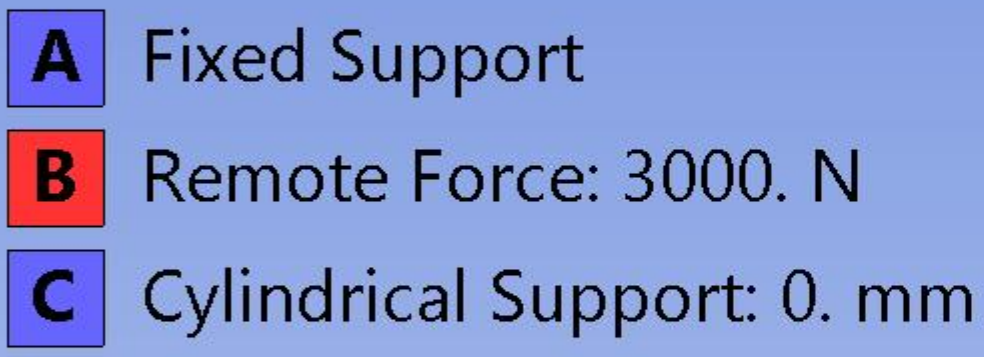
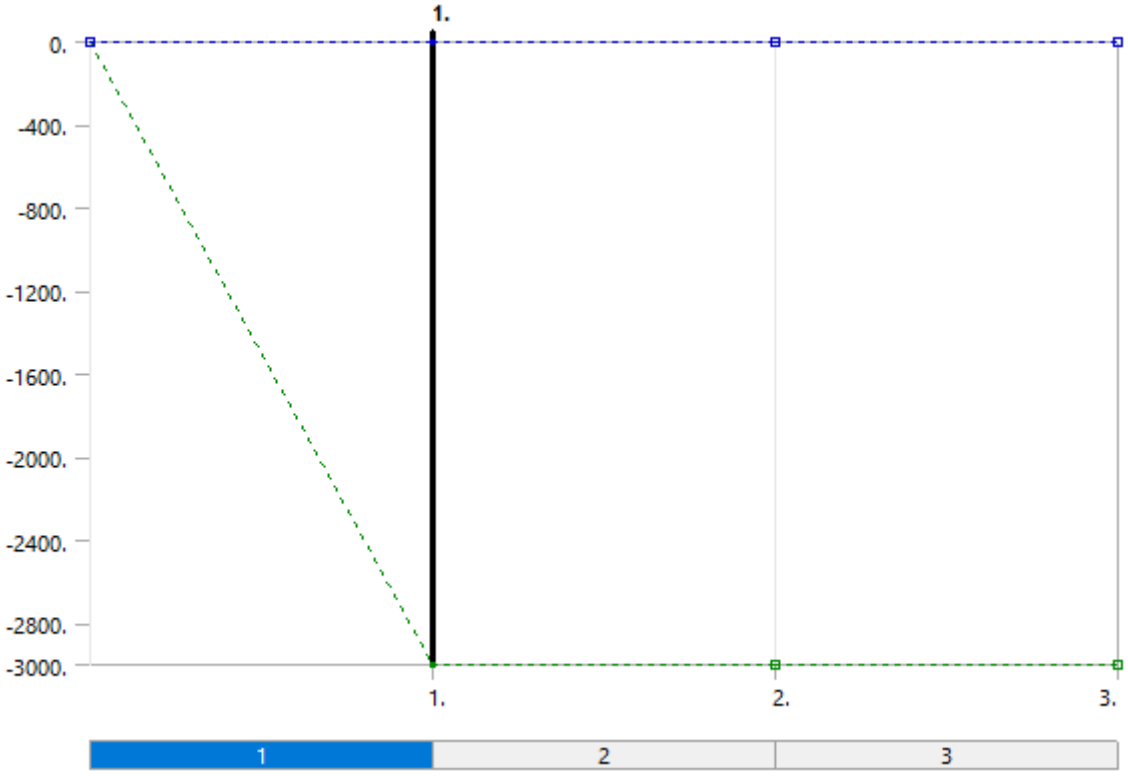


TABLE 19
Model (A4) > Static Structural (A5) > Loads

Object Name	*Fixed Support*	*Remote Force*	*Cylindrical Support*
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Geometry	2 Faces		
Coordinate System		Global Coordinate System	
X Coordinate		8.6116e-012 m	
Y Coordinate		0.14649 m	
Z Coordinate		-5.3259e-011 m	
Location		Defined	
Definition			
Type	Fixed Support	Remote Force	Cylindrical Support
Suppressed	No		
Define By		Components	
X Component		0. N (ramped)	
Y Component		-3000. N (ramped)	
Z Component		0. N (ramped)	
Behavior		Deformable	
Radial			Fixed
Axial			Free
Tangential			Fixed
Advanced			
Pinball Region		All	

FIGURE 4
Model (A4) > Static Structural (A5) > Remote Force



Solution (A6)

TABLE 20
Model (A4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	8 m 2 s
MAPDL Memory Used	5.1182 GB
MAPDL Result File Size	143.19 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 21
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Force Convergence
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

FIGURE 5
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

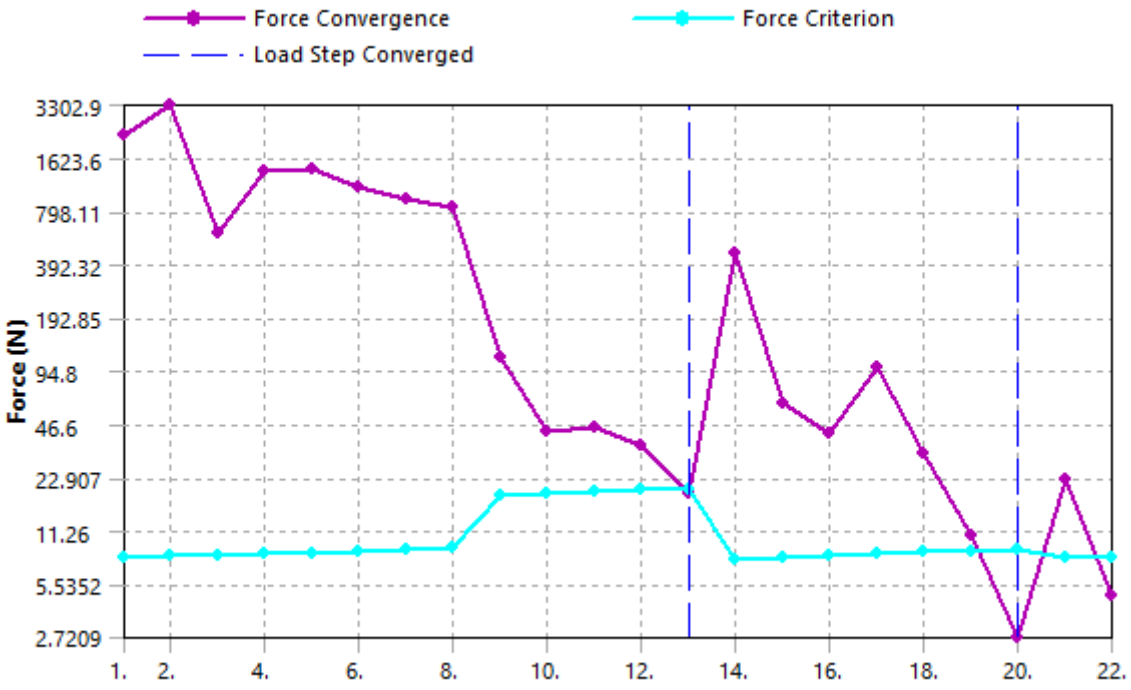


FIGURE 6
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

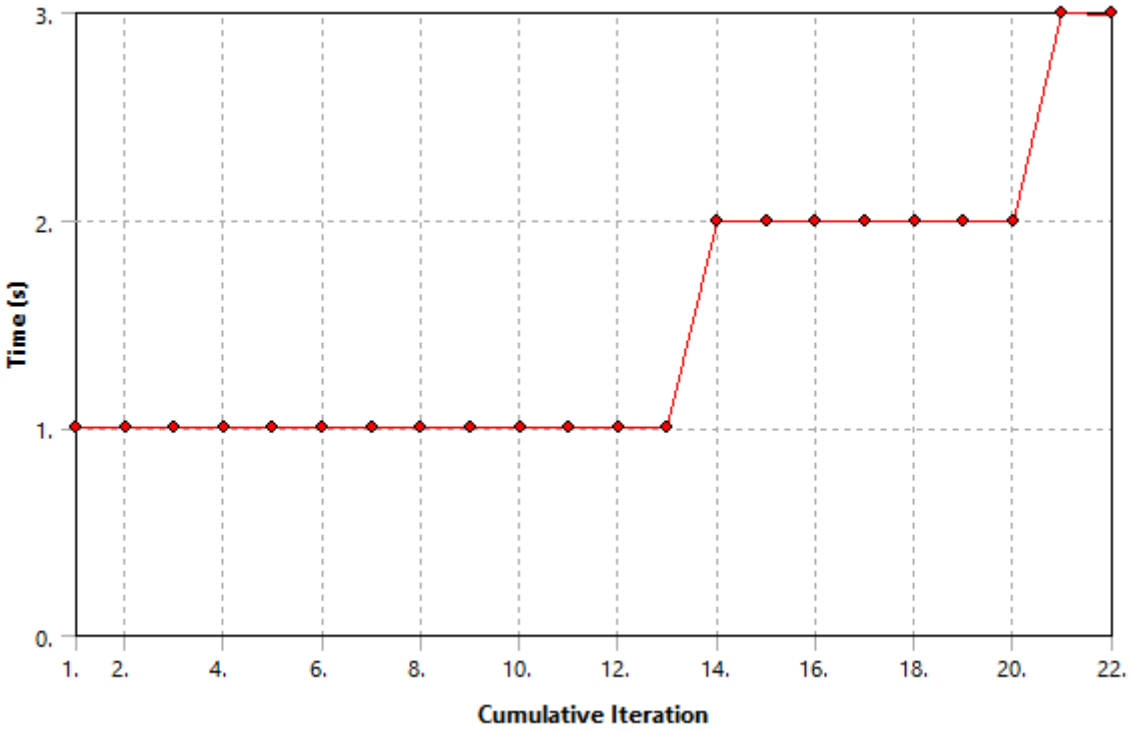


TABLE 22
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Equivalent Stress*	*Equivalent Elastic Strain*	*Total Deformation*

State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Type	Equivalent (von-Mises) Stress	Equivalent Elastic Strain	Total Deformation
By	Time		
Display Time	Last		
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Integration Point Results			
Display Option	Averaged		
Average Across Bodies	No		
Results			
Minimum	48.567 Pa	1.3197e-009 m/m	0. m
Maximum	6.7155e+007 Pa	0.10665 m/m	7.0508e-003 m
Average	2.6373e+006 Pa	8.3254e-004 m/m	1.7739e-003 m
Minimum Occurs On	STEP - Real Shock Absorber-FreeParts|Scale1[3]		STEP - Real Shock Absorber-FreeParts|Scale1[12]
Maximum Occurs On	STEP - Real Shock Absorber-FreeParts|Scale1[2]	STEP - Real Shock Absorber-FreeParts|Scale1[8]	STEP - Real Shock Absorber-FreeParts|Scale1[4]
Minimum Value Over Time			
Minimum	48.567 Pa	1.1407e-009 m/m	0. m
Maximum	71.149 Pa	1.5342e-009 m/m	0. m
Maximum Value Over Time			
Minimum	5.659e+007 Pa	0.10579 m/m	7.0508e-003 m
Maximum	6.7155e+007 Pa	0.11793 m/m	7.2101e-003 m
Information			
Time	3. s		
Load Step	3		
Substep	1		
Iteration Number	22		

FIGURE 7
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

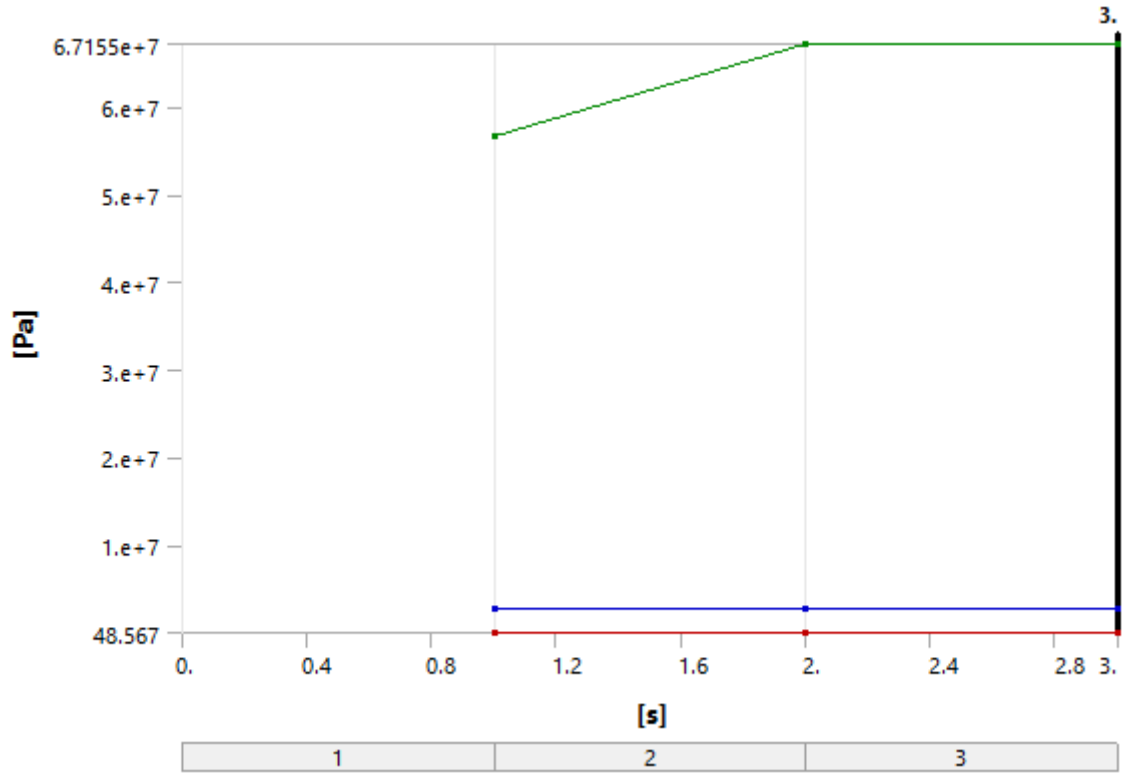


TABLE 23
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1.	71.149	5.659e+007	2.6523e+006
2.	52.496	6.7115e+007	2.6315e+006
3.	48.567	6.7155e+007	2.6373e+006

FIGURE 8
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Image

A: Static Structural

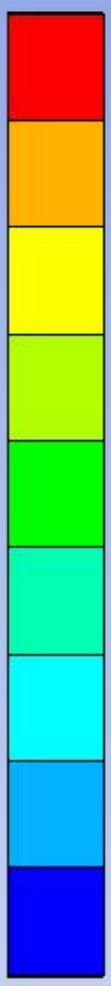
Equivalent Stress

Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 3 s

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67.155 Max

59.693

52.232

44.77

37.308

29.847

22.385

14.923

7.4617

4.8567e-5 Min

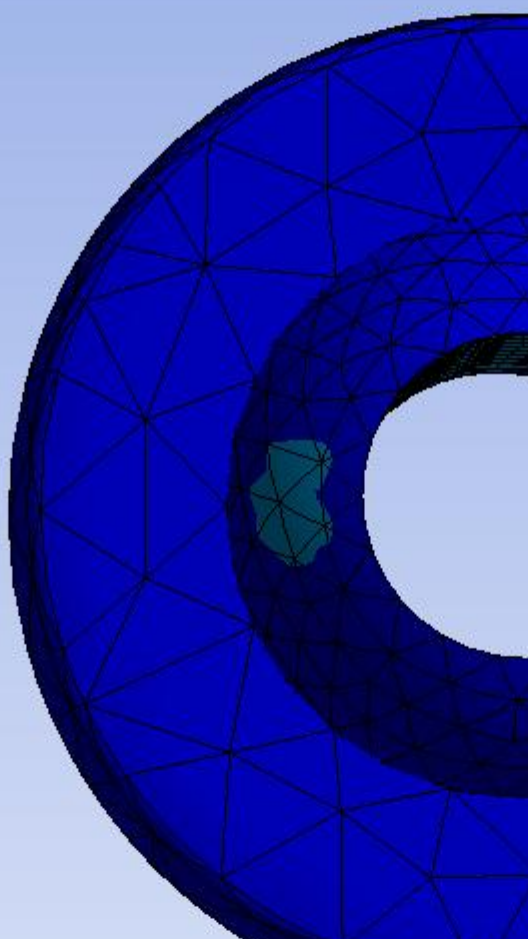


FIGURE 9
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain

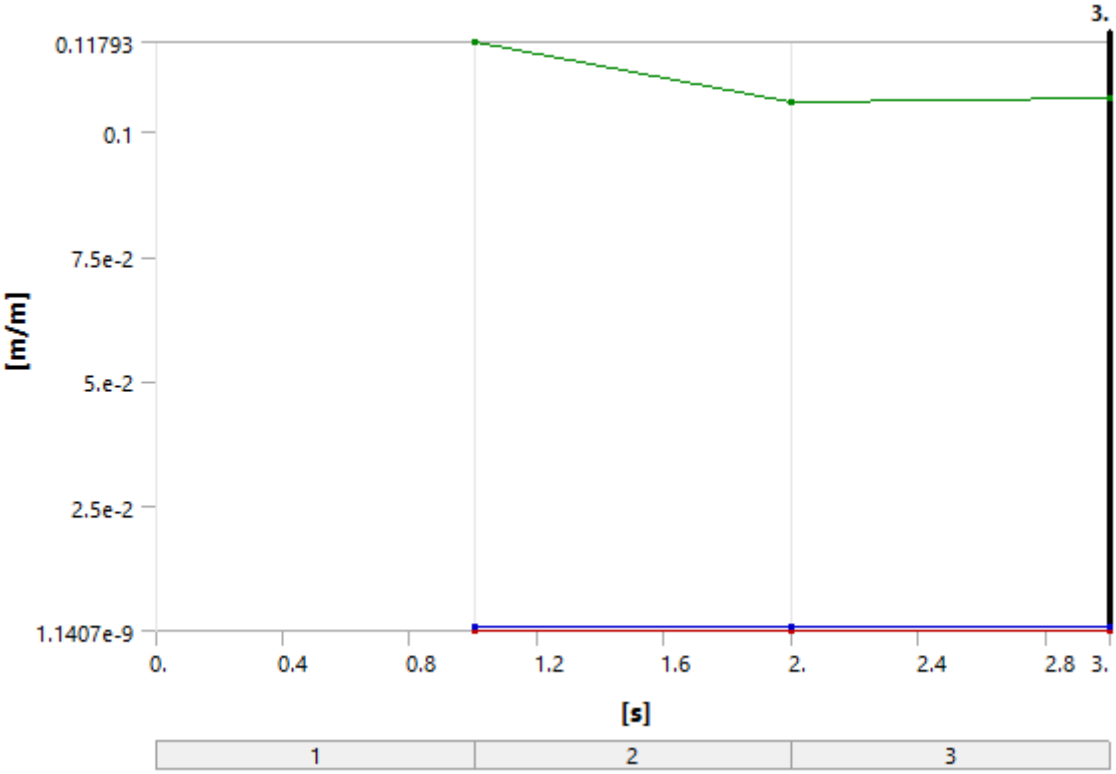


TABLE 24
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain

Time [s]	Minimum [m/m]	Maximum [m/m]	Average [m/m]
1.	1.5342e-009	0.11793	8.4402e-004
2.	1.1407e-009	0.10579	8.323e-004
3.	1.3197e-009	0.10665	8.3254e-004

FIGURE 10
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Elastic Strain > Image

A: Static Structural

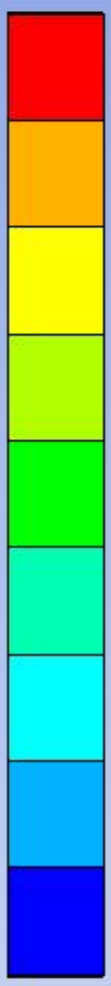
Equivalent Elastic Strain

Type: Equivalent Elastic Strain

Unit: mm/mm

Time: 3 s

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0.10665 Max

0.094803

0.082953

0.071102

0.059252

0.047402

0.035551

0.023701

0.01185

1.3197e-9 Min

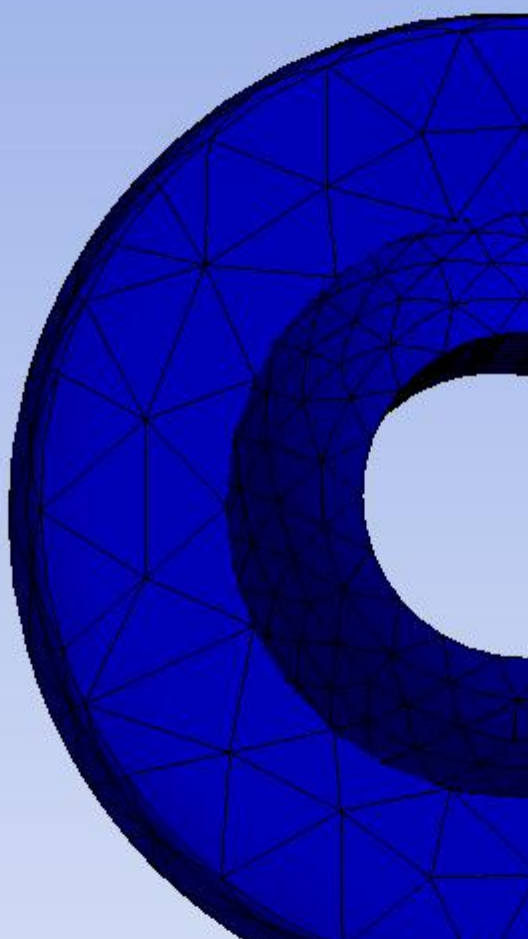


FIGURE 11
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

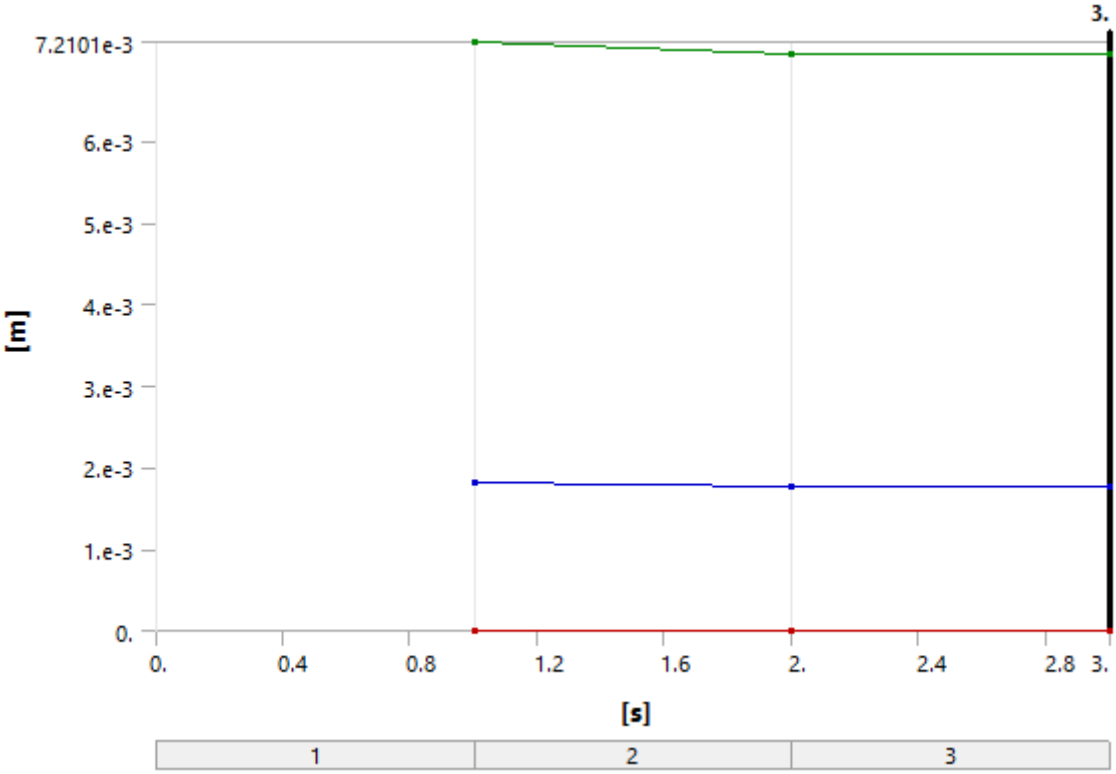


TABLE 25
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1.	0.	7.2101e-003	1.8206e-003
2.		7.0561e-003	1.7766e-003
3.		7.0508e-003	1.7739e-003

FIGURE 12
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation > Image

A: Static Structural

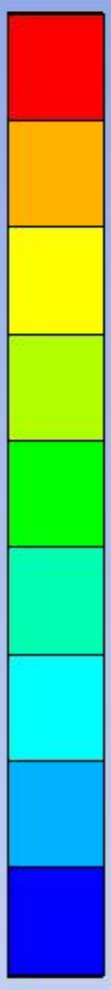
Total Deformation

Type: Total Deformation

Unit: mm

Time: 3 s

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7.0508 Max

6.2674

5.484

4.7006

3.9171

3.1337

2.3503

1.5669

0.78343

0 Min

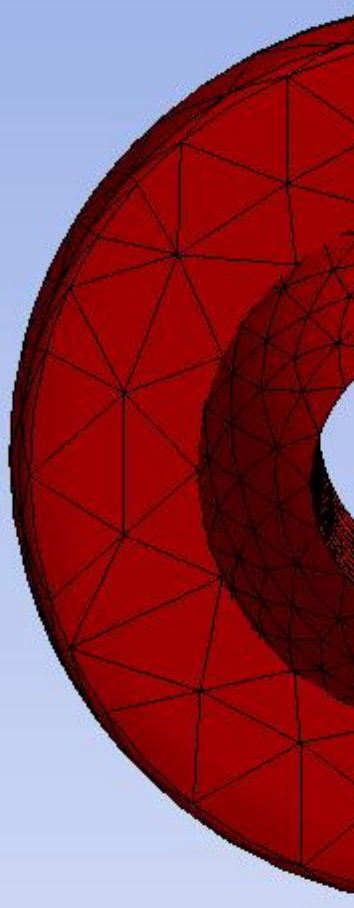


TABLE 26
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	*Stress Tool*
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 27
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results

Object Name	*Safety Factor*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	3.8597
Minimum Occurs On	STEP - Real Shock Absorber-FreeParts|Scale1[2]
Minimum Value Over Time	
Minimum	3.8597
Maximum	4.5803
Maximum Value Over Time	
Minimum	15.
Maximum	15.
Information	
Time	3. s
Load Step	3
Substep	1
Iteration Number	22

FIGURE 13
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

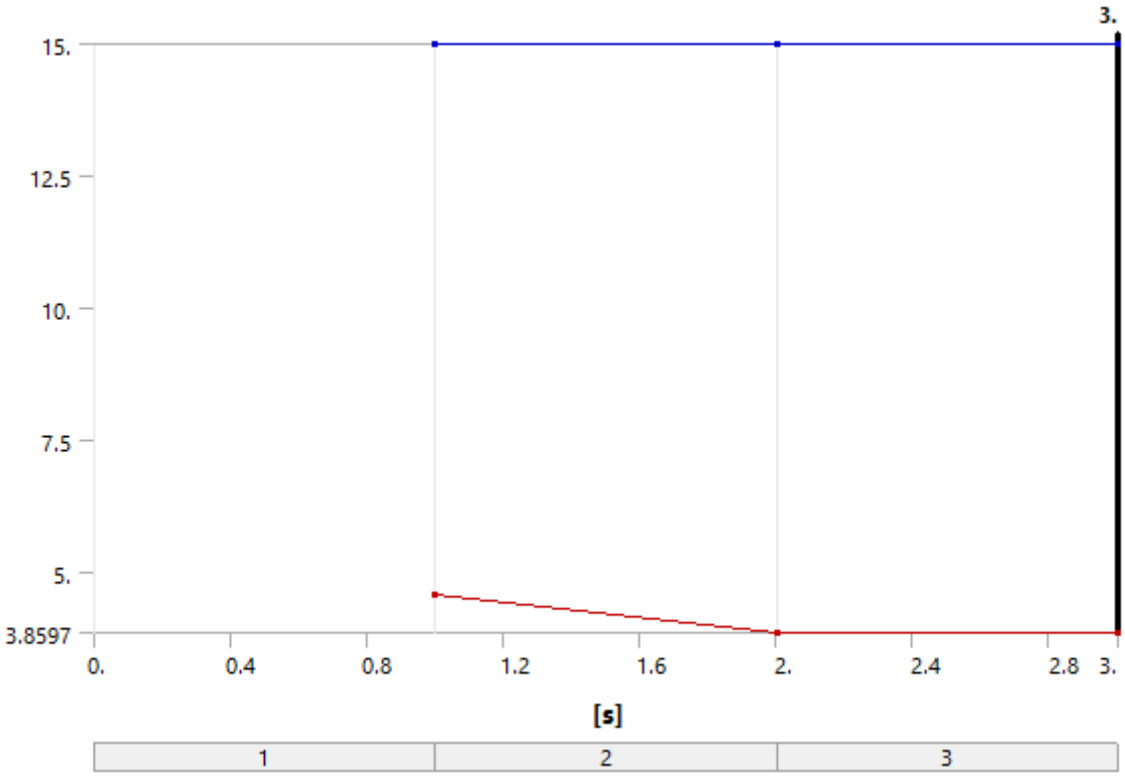


TABLE 28
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum	Average
1.	4.5803	15.	14.99
2.	3.862		14.989
3.	3.8597		

FIGURE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor > Image

A: Static Structural

Safety Factor

Type: Safety Factor

Time: 3

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15 Max

10

3.8597 Min

0

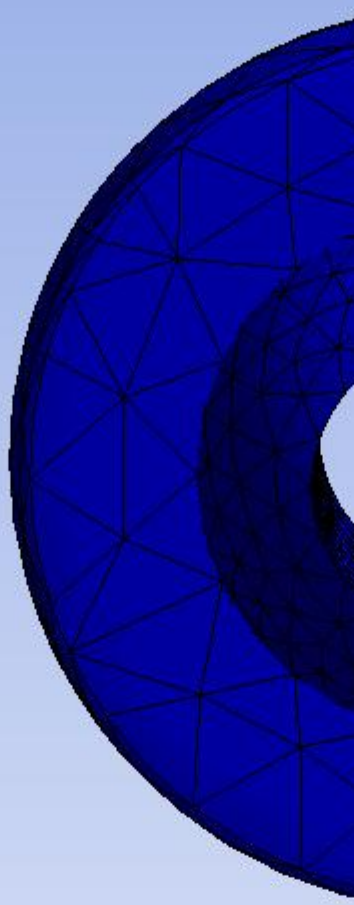


TABLE 29
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tools

Object Name	*Fatigue Tool*
State	Solved
Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Fully Reversed
Scale Factor	1.
Definition	
Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	None
Stress Component	Equivalent (von-Mises)
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

FIGURE 15
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

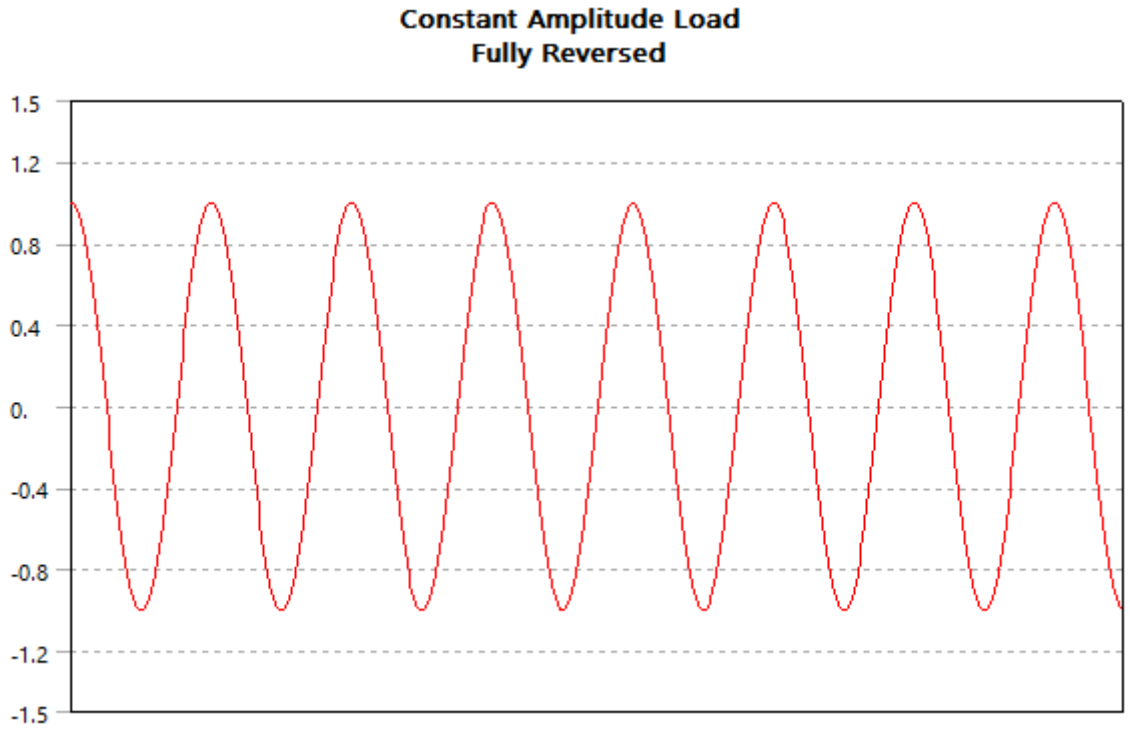


FIGURE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool

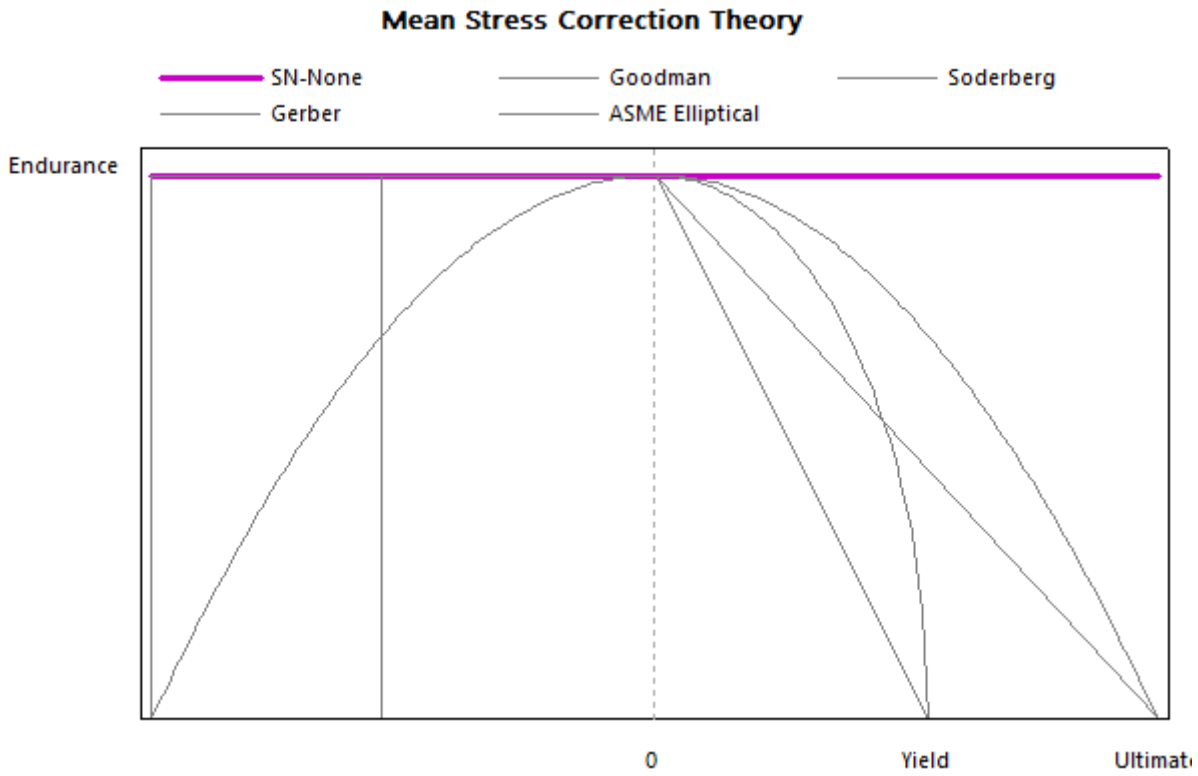


TABLE 30
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Results

Object Name	*Safety Factor*	*Life*	*Damage*
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies		
Definition			
Design Life	1.e+006 cycles		1.e+006 cycles
Type	Safety Factor	Life	Damage
Identifier			
Suppressed	No		
Results			
Minimum	1.9908	1.e+007 cycles	
Minimum Occurs On	STEP - Real Shock Absorber-FreeParts|Scale1[2]	STEP - Real Shock Absorber-FreeParts|Scale1[13]	
Maximum			0.1
Maximum Occurs On			STEP - Real Shock Absorber-FreeParts|Scale1[13]

FIGURE 17
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Safety Factor > Image

A: Static Structural

Safety Factor

Type: Safety Factor

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15 Max

10

1.9908 Min

0

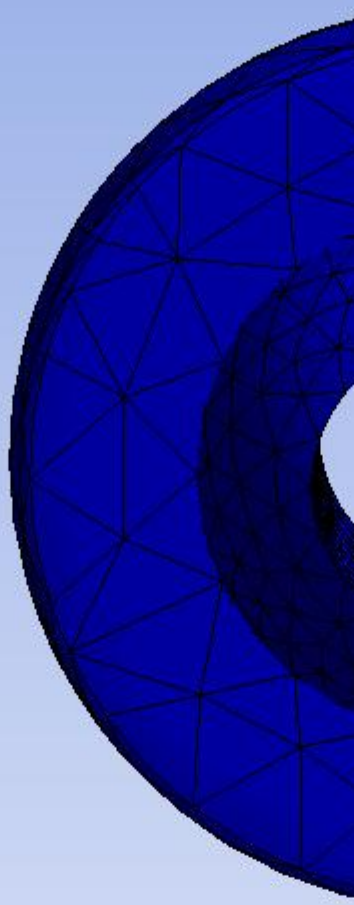


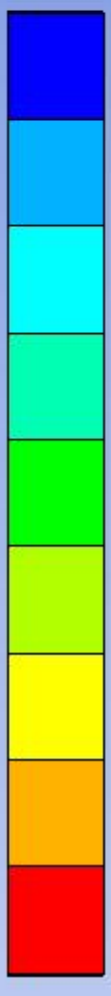
FIGURE 18
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Life > Image

A: Static Structural

Life

Type: Life

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1e8 Max

7.7426e7

5.9948e7

4.6416e7

3.5938e7

2.7826e7

2.1544e7

1.6681e7

1.2915e7

1e7 Min

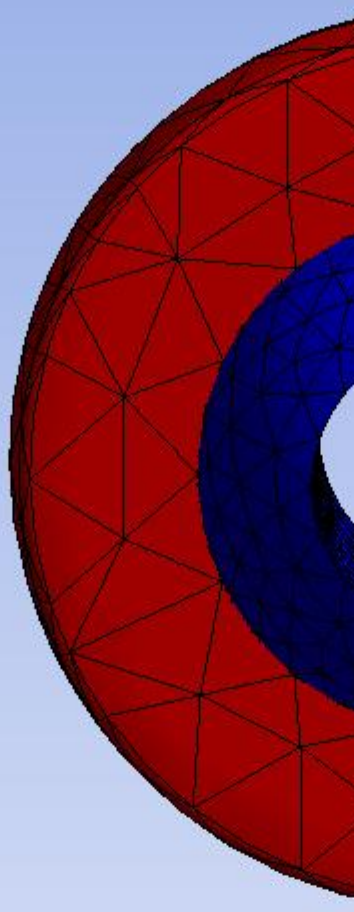


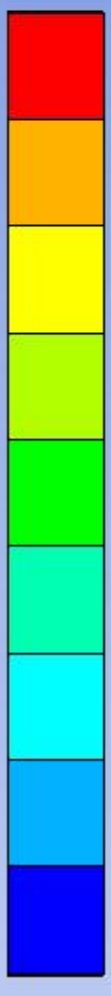
FIGURE 19
Model (A4) > Static Structural (A5) > Solution (A6) > Fatigue Tool > Damage > Image

A: Static Structural

Damage

Type: Damage

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0.1 Max

0.077426

0.059948

0.046416

0.035938

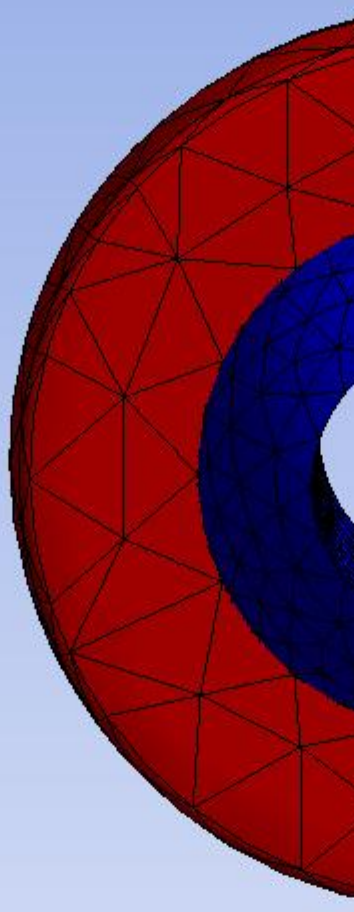
0.027826

0.021544

0.016681

0.012915

0.01 Min



Material Data

Aluminum Alloy 6061-T6

TABLE 31
Aluminum Alloy 6061-T6 > Constants

Density	2713 kg m^-3
Coefficient of Thermal Expansion	2.3e-005 C^-1
Specific Heat	875 J kg^-1 C^-1

TABLE 32
Aluminum Alloy 6061-T6 > Color

Red	Green	Blue
138	104	46

TABLE 33
Aluminum Alloy 6061-T6 > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0

TABLE 34
Aluminum Alloy 6061-T6 > Compressive Yield Strength

Compressive Yield Strength Pa
2.8e+008

TABLE 35
Aluminum Alloy 6061-T6 > Tensile Yield Strength

Tensile Yield Strength Pa
2.592e+008

TABLE 36
Aluminum Alloy 6061-T6 > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
3.131e+008

TABLE 37
Aluminum Alloy 6061-T6 > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22

TABLE 38
Aluminum Alloy 6061-T6 > Isotropic Thermal Conductivity

Thermal Conductivity W m^-1 C^-1	Temperature C
114	-100
144	0
165	100
175	200

TABLE 39
Aluminum Alloy 6061-T6 > S-N Curve

Alternating Stress Pa	Cycles	R-Ratio
2.758e+008	1700	-1
2.413e+008	5000	-1

2.068e+008	34000	-1
1.724e+008	1.4e+005	-1
1.379e+008	8.e+005	-1
1.172e+008	2.4e+006	-1
8.963e+007	5.5e+007	-1
8.274e+007	1.e+008	-1
1.706e+008	50000	-0.5
1.396e+008	3.5e+005	-0.5
1.086e+008	3.7e+006	-0.5
8.791e+007	1.4e+007	-0.5
7.757e+007	5.e+007	-0.5
7.239e+007	1.e+008	-0.5
1.448e+008	50000	0
1.207e+008	1.9e+005	0
1.034e+008	1.3e+006	0
9.308e+007	4.4e+006	0
8.618e+007	1.2e+007	0
7.239e+007	1.e+008	0
7.412e+007	3.e+005	0.5
7.067e+007	1.5e+006	0.5
6.636e+007	1.2e+007	0.5
6.205e+007	1.e+008	0.5

TABLE 40
Aluminum Alloy 6061-T6 > Isotropic Resistivity

Resistivity ohm m	Temperature C
2.43e-008	0
2.67e-008	20
3.63e-008	100

TABLE 41
Aluminum Alloy 6061-T6 > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
6.9036e+010	0.33	6.7682e+010	2.5953e+010	

TABLE 42
Aluminum Alloy 6061-T6 > Isotropic Relative Permeability

Relative Permeability
1

Low Alloy Steel 4140 - Hardened and Tempered

TABLE 43
Low Alloy Steel 4140 - Hardened and Tempered > Constants

Density	7850 kg m^-3

TABLE 44
Low Alloy Steel 4140 - Hardened and Tempered > Color

Red	Green	Blue
234	247	209

TABLE 45
Low Alloy Steel 4140 - Hardened and Tempered > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.05e+008	0.29	1.627e+008	7.9457e+007	

TABLE 46
Low Alloy Steel 4140 - Hardened and Tempered > Tensile Yield Strength

Tensile Yield Strength Pa
6.55e+008

TABLE 47
Low Alloy Steel 4140 - Hardened and Tempered > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
8.e+008

TABLE 48
Low Alloy Steel 4140 - Hardened and Tempered > S-N Curve

Alternating Stress Pa	Cycles
9.e+008	1000
7.5e+008	10000
6.e+008	1.e+005
5.e+008	1.e+006
4.8e+008	1.e+007

Carbon Steel 1095 - Hardened and Tempered

TABLE 49
Carbon Steel 1095 - Hardened and Tempered > Constants

Density	7850 kg m^-3

TABLE 50
Carbon Steel 1095 - Hardened and Tempered > Color

Red	Green	Blue
155	244	255

TABLE 51
Carbon Steel 1095 - Hardened and Tempered > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.e+011	0.29	1.5873e+011	7.7519e+010	

TABLE 52
Carbon Steel 1095 - Hardened and Tempered > Tensile Yield Strength

Tensile Yield Strength Pa
5.25e+008

TABLE 53
Carbon Steel 1095 - Hardened and Tempered > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
9.65e+008

TABLE 54
Carbon Steel 1095 - Hardened and Tempered > S-N Curve

Alternating Stress Pa	Cycles
9.e+008	1000
7.5e+008	10000

6.e+008	1.e+005
5.e+008	1.e+006
4.8e+008	1.e+007

ASTM A228 Music Wire

TABLE 55
ASTM A228 Music Wire > Constants

Density	7850 kg m^-3

TABLE 56
ASTM A228 Music Wire > Color

Red	Green	Blue
161	209	255

TABLE 57
ASTM A228 Music Wire > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
2.03e+011	0.3	1.6917e+011	7.8077e+010	

TABLE 58
ASTM A228 Music Wire > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
1.75e+009

TABLE 59
ASTM A228 Music Wire > Tensile Yield Strength

Tensile Yield Strength Pa
1.59e+009

TABLE 60
ASTM A228 Music Wire > S-N Curve

Alternating Stress Pa	Cycles
1.4e+009	1000
1.15e+009	10000
8.5e+008	1.e+005
6.5e+008	1.e+006
5.5e+008	1.e+007
5.e+008	1.e+008

Rubber, nitrile (NBR)

TABLE 61
Rubber, nitrile (NBR) > Constants

Density	984.4 kg m^-3
Tensile Yield Strength	5.292e+006 Pa
Tensile Ultimate Strength	5.292e+006 Pa
Coefficient of Thermal Expansion	2.335e-004 C^-1
Thermal Conductivity	0.1399 W m^-1 C^-1
Resistivity	1.e+010 ohm m

TABLE 62
Rubber, nitrile (NBR) > Opacity

Red	Green	Blue
153	153	204
Opacity		
0.5		
Metallic Finish		
0		

TABLE 63
Rubber, nitrile (NBR) > Mooney-Rivlin 2 Parameter

Material Constant C10 Pa	Material Constant C01 Pa	Incompressibility Parameter D1 Pa^-1	Temperature C
2.1e+005	53000	1.2e-009	23

TABLE 64
Rubber, nitrile (NBR) > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
23

TABLE 65
Rubber, nitrile (NBR) > Specific Heat Constant Pressure

Specific Heat J kg^-1 C^-1	Temperature C
458.7	-190.2
532.1	-180.2
604.3	-170.2
675.1	-160.2
744.6	-150.2
812.7	-140.2
879.6	-130.2
945.1	-120.2
1009	-110.2
1072	-100.2
1134	-90.15
1194	-80.15
1361	-70.15
1648	-60.15
1413	-50.15
1481	-40.15
1547	-30.15
1611	-20.15
1672	-10.15
1732	-0.15
1789	9.85
1844	19.85
1872	25